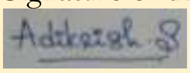
SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science & Engineering
ASSESSMENT TOOL 2 – INDUSTRY-BASED PROBLEM SOLVING

Course Code	**DSA0404**	Course Title	**Fundamentals of Data Science**
CO Assessed	**CO4 – Industry-based Problem Solving**	Assessment	**Assessment Tool 2: Industry-Based Problem Solving**
Weightage	**40% of CIA**	Total Marks	**100**
CO4 Statement	**To apply the statistical inferences such as testing of hypothesis and point estimates for the given data**		
Student Name	**Aditya Krishnan S**	Reg. No.	**192424069**
Section/Batch	**2024**	Date	**19 August 2026**

Code of Conduct:

I, ADITYA KRISHNAN S/192424069, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Industry Problem	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1. Software Industry - Confidence Interval

1. Identify population and sample

- Population: Response times of all API requests.
- Sample: 100 API requests.

2. Parameter of interest

μ = true mean API response time

3. Point estimate

$\bar{x} = \mathbf{240ms}$

4. Null hypothesis

H_0: μ = 240ms

5. Alternative hypothesis

H_1: $\mu \neq$ 240ms

6. Significance level

α = 0.05 = 5%

7. Confidence interval / test statistic

Given:
s=35, n=100, df=99

$$SE = \frac{35}{\sqrt{100}} = 3.5$$

For 95% Confidence interval, $t_{0.025,99} \approx 1.984$
CI = 240 ± 1.984(3.5) = $\boxed{(233.06, 246.94)\text{ms}}$

8. p-value

Not required for a confidence-interval-only estimation problem.

9. Compare p-value with α

Not applicable.

10. Industry-oriented conclusion

The true average API response time is estimated to lie between 233.06 ms and 246.94 ms with 95% confidence.

Question 2: E-commerce — Confidence Interval for Proportion

1. Identify population and sample

- Population: All visitors to the e-commerce platform.
- Sample: 600 randomly selected visitors.

2. Parameter of interest

p = true conversion rate

3. Point estimate

$$\hat{p} = \frac{420}{600} = 0.70$$

4. Null hypothesis
H_0: p=0.70

5. Alternative hypothesis
$H_{1:}$: p≠0.70

6. Significance level
$\alpha = 0.05 = 5\%$

7. Confidence interval / test statistic

$$\mathbf{SE} = \sqrt{\frac{\mathbf{0.70(0.30)}}{\mathbf{600}}} \approx \mathbf{0.01871}$$

CI = 0.70 ± 1.96(0.01871)
= $\boxed{(0.6633, 0.7367)}$ or $\boxed{(66.33\%, 73.67\%)}$

8. p-value
Not required for the CI estimation.

9. Compare p-value with α
Not applicable.

10. Industry-oriented conclusion
The true conversion rate is estimated to be between 66.33% and 73.67% with 95% confidence.

Question 3: Telecom — 99% Confidence Interval

1. Identify population and sample
- Population: Monthly data consumption of all telecom users.
- Sample: 225 users.

2. Parameter of interest
μ = true population mean monthly data consumption

3. Point estimate
$\boldsymbol{\bar{x} = 14.8GB}$

4. Null hypothesis
H_0: μ = 14.8GB

5. Alternative hypothesis
H_1: μ ≠ 14.8GB

6. Significance level
For 99% confidence interval,
$\alpha = 0.01$

7. Confidence interval / test statistic
Given:
s=4.5, n=225, df=224

$$\mathrm{SE} = \frac{4.5}{\sqrt{225}} = 0.3$$

For 95% Confidence interval, $t_{0.005,224} \approx 2.601$
CI = 14.8 ± 2.601(0.3) = $\boxed{(14.02, 15.58)\text{GB}}$

8. p-value
Not required for the CI estimation.

9. Compare p-value with α
Not applicable.

10. Industry-oriented conclusion
The true average monthly data consumption is estimated to lie between 14.02 GB and 15.58 GB with 99% confidence.

Question 4: Manufacturing - Hypothesis Testing Using CI

1. Identify population and sample
- Population: Diameters of all components produced by the machine.
- Sample: 36 components.

2. Parameter of interest
μ = true mean component diameter

3. Point estimate
$\bar{x} = 50.8mm$

4. Null hypothesis
H_0: μ = 50.8mm

5. Alternative hypothesis
H_1: μ ≠ 50.8mm

6. Significance level
α = 0.05 = 5%

7. Confidence interval / test statistic
Given,
s = 2.4, n = 36, df = 35

$$SE = \frac{2.4}{\sqrt{36}} = 0.4$$

Test Statistic,

$$t = \frac{50.8 - 50}{0.4} = \boxed{2.00}$$

For 95% CI:
$t_{0.025,\,35} \approx \boxed{2.030}$
CI = 50.8 ± 2.030(0.4)
$\boxed{(49.99, 51.61)\ mm}$

8. p-value
For t=2.00, df=35
$\boxed{p \approx 0.053}$

9. Compare p-value with α
0.053>0.05

Therefore:
$\boxed{\boldsymbol{Failed\ to\ reject\ H_0}}$

10. Industry-oriented conclusion
Since 50 mm lies within the 95% confidence interval and (p>0.05), there is insufficient evidence to conclude that the machine's true mean diameter differs from 50 mm.

Question 5: Banking — Hypothesis Testing Using CI

1. Identify population and sample
- Population: Loan-processing times for all applications.
- Sample: 49 loan applications.

2. Parameter of interest
μ = true mean loan processing time

3. Point estimate
$\bar{x} = 5.6\ days$

4. Null hypothesis
H_0: μ = 5 days

5. Alternative hypothesis
H_1: μ ≠ 5 days

6. Significance level
α = 0.05 = 5%

7. Confidence interval / test statistic
Given,
s = 1.4, n = 49, df = 48

$$SE = \frac{1.4}{\sqrt{49}} = 0.2$$

Test Statistic,
$$t = \frac{5.6 - 5}{0.2} = \boxed{3.00}$$

For 95% CI:
$t_{0.025,\,48} \approx \boxed{2.011}$

CI = 50.8 ± 2.011(0.2)
= $\boxed{(5.20, 6.00)\ days}$

8. p-value
For t=3.00, df=48
$\boxed{p \approx 0.004}$

9. Compare p-value with α
0.004>0.05

Therefore:
$\boxed{\boldsymbol{Reject\ H_0}}$

10. Industry-oriented conclusion
Since 50 mm lies within the 95% confidence interval and ($p>0.05$), there is insufficient evidence to conclude that the machine's true mean diameter differs from 50 mm.
